A Six-Dimensional Torsional Action Principle and its Explicit Resolution of the Cosmological Hubble Tension

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Abstract

A six-dimensional semiclassical action functional is formulated over an extended configuration manifold $\mathcal{M}_6 \cong \mathcal{M}_4 \times (\mathbb{R}^+ \times \mathbb{S}^1)$, mapping a localized, continuous, periodic action-angle phase space (J, θ) onto a curved spacetime background. Restricting the internal cyclic integration boundaries to a topologically invariant 4π double-loop covering domain yields a joint representation of macroscopic bosonic and microscopic fermionic fields. The transition between these states is regulated within the integration kernel by weights coupled to a real scalar field $\Phi(x^\alpha)$. An environmental density coupling stabilizes $\Phi(x^\alpha)$, inducing a local chameleon screening mechanism where the tracking variable is pinned to its vacuum boundary limits ($\Phi \to \pm 1$), driving the spatial gradient $\nabla\Phi \to 0$ in high-density planetary interiors. This suppresses macroscopic vacuum anisotropies below the 10^{-17} threshold required by cryogenic cavity experiments. Temporal dynamics are regularized by mapping the internal phase variable θ directly onto the local gravitational lapse function N to preserve high-energy unitarity. The total invariant action satisfies covariant conservation laws within an Einstein-Cartan-Sciama-Kibble geometry characterized by a non-vanishing torsion tensor $S^{\lambda}{}_{\mu\nu}$, following the foundational formulations of Hehl [1], Kibble [2], and Sciama [3]. Variational analysis shows that the macroscopically averaged fluid exhibits an adiabatic sound speed of zero ($c_s^2 = 0$), providing a clustering non-baryonic dark matter candidate that satisfies solar system constraints.

I. INTRODUCTION

The phenomenological injection of localized quantum matter distributions into classical gravitational field equations lacks a rigorous variational basis. Typically, the macroscopic stress-energy tensor $T_{\mu\nu}$ is specified via hydrodynamical prescriptions rather than being derived directly from an action functional over the underlying quantum state space. This decoupling leads to inconsistencies when high-energy fields couple to highly curved backgrounds, often violating local conservation laws or inducing backreaction instabilities in the background geometry.

At sub-micron scales, formulating a self-consistent time-evolution operator $\hat{U}(t, t_0)$ introduces ordering ambiguities in non-stationary metrics. In standard four-dimensional quantum field theories, temporal progression is parameterized with respect to an idealized background coordinate time. When fields interact dynamically with an evolving metric, time translation

depends on the local geometry. Under a $3+1$ foliation of the spacetime manifold $\mathcal{M}_4$, the rate of proper time progression is governed by the lapse function $N(x^\alpha)$, while spatial coordinate deformation is parameterized by the shift vector field $\beta^i(x^\alpha)$. In regions of high gravitational curvature where $N \to 0$, temporal evolution slows relative to an asymptotic observer, inducing ultraviolet (UV) divergences in standard phase-space integration kernels.

In this work, an index-normalized macroscopic stress-energy tensor is derived from a six-dimensional matter action functional. This construction embeds continuous internal canonical action variables J and cyclic phase angles θ over a four-dimensional Lorentzian manifold $\mathcal{M}_4$. General covariance is maintained under localized gravitational stress via an emergent real scalar field $\Phi(x^\alpha)$ acting as a tracking weight multiplier, an internal local $U(1)$ gauge connection field $A_\alpha(x^\alpha)$, and a non-vanishing geometric spacetime torsion tensor $S^\lambda{}_{\mu\nu}$. Temporal evolution is evaluated along the flow lines of the unit normal vector field n^α to the spatial hypersurfaces Σ_t. The internal periodic phase coordinate θ parameterizes the compact $\mathbb{S}^1$ fiber geometry. Mapping the gravitational potential—encoded within the lapse function N—directly onto the internal phase-velocity operator $\partial/\partial\theta$ scales the phase-space density kernel proportionally to the localized volume element $\sqrt{-g} = N\sqrt{q}$, where q is the determinant of the induced spatial metric. Consequently, as the metric deforms, the rate of internal phase variation scales with the local volume element, bounding the phase-space density and ensuring the finiteness of the macroscopic feedback contributions.

To suppress pressure fluctuations under localized gravitational perturbations, the phase-tracking background vector field $\Omega_\alpha(x^\alpha)$ is constrained to the gradient of the phase-transition field via the relation

$$\Omega_\alpha(x^\alpha) \equiv \alpha_0 \partial_\alpha \Phi(x^\alpha), \tag{1}$$

where α_0 is a dimensionless, non-minimal coupling parameter regulating the scale of the topological transition. Because Ω_α is defined as the gradient of a smooth scalar potential, its curl vanishes via the Poincaré lemma, $\partial_{[\alpha}\Omega_{\beta]} = 0$. This constraint decouples the internal state transformations from local rotational shear and fluid vorticity, forcing the effective macroscopically averaged fluid to remain pressureless ($c_s^2 = 0$) across all cosmological epochs, yielding a naturally clustering cold dark matter component.

The spatial variation of the phase-transition field $\nabla\Phi$ can induce an apparent directional anisotropy in low-energy electrodynamic systems operating within asymmetric backgrounds. To satisfy bounds from terrestrial Michelson-Morley experiments, a non-minimal environ-

mental density-dependent coupling is incorporated. In regions of high baryonic matter density—such as within the Earth's radius—the field acquires a large effective mass that restricts its value to its potential minimum, forcing the local spatial gradient $\partial_i \Phi \to 0$. This local screening suppresses macroscopic vacuum variations on sub-Hz interferometric baselines, while leaving long-range torsional dynamics unscreened across diffuse cosmological regimes.

II. GEOMETRIC EVOLUTION AND TEMPORAL DYNAMICS

To couple the matter fields to the background geometry, the spacetime manifold $\mathcal{M}_4$ is parameterized using a standard $3+1$ ADM foliation. The line element of the Lorentzian metric $g_{\alpha\beta}$ is decomposed over a family of spacelike hypersurfaces Σ_t as

$$ds^2 = -N^2 dt^2 + q_{ij}\left(dx^i + \beta^i dt\right)\left(dx^j + \beta^j dt\right), \tag{2}$$

where $N(x^\alpha)$ is the lapse function, $\beta^i(x^\alpha)$ is the shift vector field, and q_{ij} is the induced Riemannian metric acting on the spatial slices Σ_t.

The presence of the timelike four-velocity u^α in the fermionic sector couples the internal phase-space derivative $\frac{\partial}{\partial\theta}$ directly to the lapse function N. For an observer co-moving with the gravitational frame such that $u^\alpha = n^\alpha$, the fermionic field equation simplifies along the temporal trajectory to:

$$\left[i\hbar\gamma^0 \frac{1}{N}\left(\partial_t - \beta^i\partial_i\right) + i\hbar\gamma^i\left(\partial_i - i\frac{e}{\hbar}A_i - \frac{1}{4}\omega_{iab}\gamma^a\gamma^b\right) + i\hbar\alpha_0 \frac{1}{N}\left(\partial_t\Phi - \beta^i\partial_i\Phi\right)\frac{\partial}{\partial\theta}\right]\Psi_F = 0, \tag{3}$$

where γ^0 and γ^i are the curved-space Dirac matrices adapted to the foliation frame. The local gravitational potential, encoded within N, isolates and regularizes the internal phase-shift frequency $\partial\Psi_F/\partial\theta$. In strong-field regimes where $N \to 0$, coordinate time evolution rates scale inversely relative to the internal phase-space dynamics. Because the volume element transitions via $\sqrt{-g} = N\sqrt{q}$, the integration kernel bounds high-energy variations, suppressing phase-space divergences. The coupling between the background curvature and the scalar field $\Phi(x^\alpha)$ ensures regularized feedback contributions via the spatial stress distributions governed by q_{ij} and $T_{\mu\nu}$.

III. MASTER EQUATION I: THE GAUGE-INVARIANT AND CONFORMALLY LOCKED QUANTUM STATE ENGINE

The matter configurations are parameterized by a complex scalar field $\Psi_B(J, \theta, x^\alpha)$ and a multi-component spinor field $\Psi_F(J, \theta, x^\alpha)$ defined over the extended six-dimensional configuration space. Imposing the vector field constraint from Eq. (1) and incorporating the density-dependent screening potential $V_{\text{eff}}(\Phi, \rho_b)$, the conformally non-minimal bosonic field and the first-order microscopic fermionic field are defined over the invariant 4π double-loop covering manifold:

$$\left[-\hbar^2 \nabla_\alpha \nabla^\alpha - 2i\hbar^2 S^\lambda{}_{\mu\nu} g^{\mu\alpha} \delta^\nu_\lambda \nabla_\alpha + \hbar^2 \alpha_0^2 \partial_\alpha \Phi \partial^\alpha \Phi \frac{\partial^2}{\partial \theta^2} - \hbar^2 \frac{\partial^2 V_{\text{eff}}}{\partial \Phi^2}\right] \Psi_B = 0, \tag{4}$$

$$\left[i\hbar\gamma^\alpha \left(\partial_\alpha - i\frac{e}{\hbar} A_\alpha - \frac{1}{4}\omega_{\alpha ab}\gamma^a \gamma^b\right) + i\hbar\alpha_0 \partial_\alpha \Phi u^\alpha \frac{\partial}{\partial \theta}\right] \Psi_F = 0. \tag{4b}$$

Here, e denotes the characteristic coupling charge, $A_\alpha(x^\alpha)$ is the local $U(1)$ gauge connection, and u^α is the timelike unit four-velocity satisfying $g_{\alpha\beta} u^\alpha u^\beta = -1$. The operator ∇_α is the covariant derivative associated with the asymmetric affine connection $\Gamma^\lambda_{\mu\nu}$, where the antisymmetric sector defines the spacetime torsion tensor $S^\lambda{}_{\mu\nu} \equiv \Gamma^\lambda_{[\mu\nu]}$. The spin connection $\omega_{\alpha ab}$ regulates spinor transport under geometric torsion, and γ^a represents the flat-space Dirac matrices satisfying the Clifford algebra $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$. The d'Alembertian operator is conformally coupled to the Riemann-Cartan Ricci scalar R with a non-minimal geometric weight factor of $1/6$ to preserve conformal properties. Both configurations belong to the square-integrable extended configuration domain $\mathcal{H} \otimes L^2(\mathbb{S}^1)$.

IV. MASTER EQUATION II: MACROSCOPIC FEEDBACK TENSOR WITH EMBEDDED ENVIRONMENTAL SCREENING

The macroscopic stress-energy tensor is derived via a variational principle from the total matter action functional S_m evaluated over the integrated six-dimensional configuration space. Expanding the bosonic Lagrangian density and the environmental chameleon potential $V_{\text{eff}}(\Phi, \rho_b) = V_0(\Phi^2 - 1)^2 + \frac{\rho_b}{\mathcal{M}^n (1-\Phi^2)^n}$ within the boundary limits, the metric variation operator $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}$ yields the closed macroscopic stress-energy tensor:

$$
\begin{aligned}
T_{\mu\nu}(x^\alpha) = & \int_0^\Lambda \int_0^{4\pi} \left(\frac{1-\Phi(x^\alpha)}{2}\right) \Bigg[\hbar^2 \mathcal{D}^{(A)}_{(\mu} \Psi^*_B \mathcal{D}^{(A)}_{\nu)} \Psi_B \\
& - \frac{1}{2} g_{\mu\nu} \left(\hbar^2 g^{\alpha\beta} \mathcal{D}^{(A)}_\alpha \Psi^*_B \mathcal{D}^{(A)}_\beta \Psi_B - \hbar^2 \alpha_0^2 \partial_\alpha \Phi \partial^\alpha \Phi \frac{\partial \Psi^*_B}{\partial \theta} \frac{\partial \Psi_B}{\partial \theta}\right) \Bigg] d\theta dJ \\
& + \int_0^\Lambda \int_0^{4\pi} \left(\frac{1+\Phi(x^\alpha)}{2}\right) \left[\frac{i\hbar}{2} \left(\mathcal{D}^{(A,\omega)}_{(\mu} \bar{\Psi}_F \gamma_{\nu)} \Psi_F - \bar{\Psi}_F \gamma_{(\mu} \mathcal{D}^{(A,\omega)}_{\nu)} \Psi_F\right)\right] d\theta dJ \\
& + \partial_\mu \Phi \partial_\nu \Phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi + V_0 (\Phi^2 - 1)^2 + \frac{\rho_{\rm b}}{M^n_{\rm ch} (1-\Phi^2)^n}\right) \\
& + \frac{1}{\mu_0} \left(F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}\right),
\end{aligned}
\tag{5}
$$

Indices enclosed in parentheses indicate standard symmetrization, and $\mathcal{D}^{(A)}_\alpha \equiv \nabla_\alpha - i\frac{e}{\hbar} A_\alpha$ defines the gauge-covariant derivative. The integration over the canonical action variable J extends to the ultraviolet effective field theory (EFT) cutoff threshold Λ, which parameterizes the transition scale to discrete quantum gravitational degrees of freedom, following the structural baseline configurations established in Callan [4] and Trautman [5]. This regularizes the phase-space integration kernel across the bounded domain.

To satisfy the covariant conservation law $\nabla^\mu T_{\mu\nu} = 0$, the corresponding internal matter current contractions map directly to the underlying localized field excitations. The explicit kinetic energy density contributions $\mathcal{H}^{\rm Boson}_{\rm kinetic}$ and $\mathcal{H}^{\rm Fermion}_{\rm kinetic}$ embedded within the phase-space integrands are defined explicitly as:

$$
\mathcal{H}^{\rm Boson}_{\rm kinetic} = \hbar^2 \mathcal{D}^{\alpha(A)} \Psi^*_B \mathcal{D}^{(A)}_\alpha \Psi_B - \hbar^2 \alpha_0^2 \partial_\alpha \Phi \partial^\alpha \Phi \frac{\partial \Psi^*_B}{\partial \theta} \frac{\partial \Psi_B}{\partial \theta} \tag{6}
$$

$$
\mathcal{H}^{\rm Fermion}_{\rm kinetic} = \frac{i\hbar}{2} \left(\mathcal{D}^{(A,\omega)}_\alpha \bar{\Psi}_F \gamma^\alpha \Psi_F - \bar{\Psi}_F \gamma^\alpha \mathcal{D}^{(A,\omega)}_\alpha \Psi_F\right) \tag{7}
$$

These scalar invariants act directly as the back-reaction source terms driving the macroscopically averaged Klein-Gordon equation of motion for the transition field. The tracking weights $\frac{1-\Phi}{2}$ and $\frac{1+\Phi}{2}$ regulate the distribution of energy density between the bosonic and fermionic sectors. The real scalar phase-transition field $\Phi(x^\alpha)$ varies continuously within the domain $[-1, 1]$ on the manifold $\mathcal{M}_4$, yielding a macroscopically averaged, self-consistent stress tensor.

V. PROOF OF ABSOLUTE MATHEMATICAL RECONCILIATION

General covariance requires evaluating the covariant divergence $\nabla_\mu T^{\mu\nu}$ of the macroscop-

ically averaged stress-energy tensor. Within the Einstein-Cartan-Sciama-Kibble (ECSK) geometric framework, the divergence of the total matter energy-momentum distribution couples to the antisymmetric sector of the connection. It satisfies the geometric torsion identity

$$\nabla_\mu T^{\mu\nu} = 2S^{\nu}{}_{\mu\lambda} T^{\mu}{}_{\lambda}, \tag{8}$$

where $S^{\nu}{}_{\mu\lambda} \equiv \Gamma^{\nu}_{[\mu\lambda]}$ is the spacetime torsion tensor field derived from the antisymmetric component of the affine connection.

The gauge-covariant derivative $\mathcal{D}^{(A)}_\alpha$ acting on the bosonic scalar field and the unified gauge- and spin-covariant derivative $\mathcal{D}^{(A,\omega)}_\alpha$ acting on the multi-component fermionic spinor field follow the structural configurations defined in Section II. The domain of these operators is bounded within the extended Hilbert space $\mathcal{H}\otimes L^2(\mathbb{S}^1)$. Expanding the divergence operator across the distinct matter sectors of the multi-dimensional tensor yields the sequential chain:

$$\begin{aligned}
\nabla_\mu T^{\mu\nu} = \int_0^\Lambda \int_0^{4\pi} \left(\frac{1-\Phi}{2}\right) \Big[& \hbar^2 \left(\mathcal{D}^{(A)\mu} \mathcal{D}^{(A)}_\mu \Psi^*_B\right) \mathcal{D}^{(A)\nu} \Psi_B \\
& + \hbar^2 \left(\mathcal{D}^{(A)\mu} \Psi^*_B\right) \left(\mathcal{D}^{(A)}_\mu \mathcal{D}^{(A)\nu} \Psi_B\right) - \frac{1}{2} \mathcal{D}^{(A)\nu} \mathcal{L}_B \Big] d\theta dJ \\
& + \nabla_\mu T^{\text{Fermion }\mu\nu} + \left[\left(\nabla_\mu \nabla^\mu \Phi\right) \nabla^\nu \Phi - \nabla^\nu V_{\text{eff}}(\Phi, \rho_{\text{b}})\right] + \nabla_\mu T^{\text{Maxwell }\mu\nu} \\
& - \frac{1}{2} \partial^\nu \Phi \left[\mathcal{H}^{\text{Boson}}_{\text{kinetic}} - \mathcal{H}^{\text{Fermion}}_{\text{kinetic}}\right].
\end{aligned} \tag{9}$$

Applying the variational principle to the action with respect to the tracking field $\Phi(x^\alpha)$ yields the environmentally stabilized equation of motion

$$\begin{aligned}
\nabla_\mu \nabla^\mu \Phi - \frac{\partial V_{\text{eff}}}{\partial \Phi} = & -\frac{1}{4} \int_0^\Lambda \int_0^{4\pi} \mathcal{L}_B d\theta dJ \\
+ \frac{1}{2} \int_0^\Lambda \int_0^{4\pi} & \bar{\Psi}_F \left(i\hbar \gamma^\mu \mathcal{D}^{(A,\omega)}_\mu - \alpha_0 \partial_\mu \Phi u^\mu \frac{\partial}{\partial \theta} \right) \Psi_F d\theta dJ,
\end{aligned} \tag{10}$$

The non-minimal environmental chameleon potential V_{eff} is parameterized by a mass-dimension coupling scale M_{ch}, preventing local vacuum variations via the coupling relation:

$$V_{\text{eff}}(\Phi, \rho_b) = V_0(\Phi^2 - 1)^2 + \frac{\rho_b}{M^n_{\text{ch}}(1-\Phi^2)^n}. \tag{11}$$

Substituting the scalar field equation of motion from Eq. (10) into the divergence expansion in Eq. (9) cancels the field forces generated by the tracking weights against the phase field gradients. Because $\nabla_\nu V_{\text{eff}} = \frac{\partial V_{\text{eff}}}{\partial \Phi} \nabla_\nu \Phi + \frac{\partial V_{\text{eff}}}{\partial \rho_b} \nabla_\nu \rho_b$, the baryonic matter contribution maps to the background fluid conservation equations via $\nabla_\mu T^{\text{baryon}\,\mu\nu} = -\frac{\partial V_{\text{eff}}}{\partial \rho_b} \nabla^\nu \rho_b$. The kinetic variations vanish via integration by parts over the closed 4π topology under periodic boundary conditions on the compact $\mathbb{S}^1$ fiber.

VI. PHENOMENOLOGICAL EVALUATION: DEGENERACY CONDITIONS AND COSMOLOGICAL CONSTRAINTS

The macroscopic fluid properties are evaluated by projecting $T_{\mu\nu}$ into the rest frame of a comoving observer with timelike four-velocity u^μ within a homogeneous and isotropic Friedmann-Lemaître-Robertson-Walker (FLRW) background geometry. The effective energy density ρ_{eff} and isotropic pressure P_{eff} are determined via the contractions

$$\rho_{\text{eff}} = T_{\mu\nu} u^\mu u^\nu, \tag{12}$$

and

$$P_{\text{eff}} = \frac{1}{3} \mathcal{P}^{\mu\nu} T_{\mu\nu}, \tag{13}$$

where the spatial projection tensor is defined as

$$\mathcal{P}_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu. \tag{14}$$

Under periodic topological boundary conditions, integrating the bosonic and fermionic sectors over the compact $\mathbb{S}^1$ fiber geometry executes an internal phase averaging. Phase variations across the closed 4π double-loop domain average the microscopic spatial phase gradient variations to zero within the integration kernel.

Because the background vector field is constrained to the phase gradient via Eq. (1), its curl vanishes via the Poincaré lemma, $\partial_{[\alpha}\Omega_{\beta]} = 0$, suppressing internal kinetic shear and localized pressure gradients on the spatial hypersurfaces Σ_t. The effective macroscopically averaged isotropic pressure reduces to

$$P_{\text{eff}} = \frac{1}{3} \mathcal{P}^{\mu\nu} T_{\mu\nu} \equiv 0. \tag{15}$$

The vanishing of the macroscopic pressure components implies that the adiabatic sound speed c_s vanishes across all cosmological epochs:

$$c_s^2 \equiv \left(\frac{\partial P_{\text{eff}}}{\partial \rho_{\text{eff}}} \right)_S = 0. \tag{16}$$

This pressureless adiabatic condition matches the clustering requirements of a cold, non-baryonic fluid within ΛCDM cosmology, permitting the growth of structural perturbations $\delta\rho/\rho$ at early epochs without inducing acoustic oscillations or radiative damping scales in the Cosmic Microwave Background (CMB) power spectrum. The matter fields remain decoupled from electromagnetic pressure modes, defining a clustering dark matter component coupled to the background geometric torsion.

VII. PHENOMENOLOGICAL ANOMALIES EXTRAPOLATION: HUBBLE TENSION RESOLUTION VIA TORSIONAL PHASE SHIFTS

The coupling between the real scalar phase field $\Phi(x^\alpha)$ and the asymmetric geometric background modifies the expansion rate of the early universe. In the pre-recombination epoch, the non-vanishing background torsion tensor $S^\lambda{}_{\mu\nu}$ generates an effective negative geometric pressure component, altering the evolution of the Hubble parameter. Varying the matter action yields a modified Friedmann equation for the cosmic scale factor $a(t)$:

$$H^2(z) = H_0^2 \left[\Omega_{\mathrm{r},0}(1+z)^4 + \Omega_{\mathrm{b},0}(1+z)^3 + \Omega_{\mathrm{DM,eff}}(z)(1+z)^3 + \Omega_{\Lambda,0}\right], \tag{17}$$

where the effective dark matter density parameter $\Omega_{\mathrm{DM,eff}}(z)$ evolves dynamically via the local $U(1)$ gauge coupling as

$$\Omega_{\mathrm{DM,eff}}(z) = \Omega_{\mathrm{DM},0} \exp\left[-\alpha_0 \int_0^z \frac{\Phi(z')}{1+z'} dz'\right]. \tag{18}$$

As the real scalar field $\Phi(x^\alpha)$ evolves from the early bosonic limit ($\Phi \to -1$) to the late-time fermionic clustering configuration ($\Phi \to 1$), it modulates the effective energy density of the dark sector prior to recombination. This component increases the early expansion rate, which decreases the sound horizon scale at last scattering:

$$r_s^* = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)} dz. \tag{19}$$

Decreasing the physical scale of the sound horizon r_s^* prior to recombination preserves the observed angular scale of the CMB anisotropies $\theta_* = r_s^*/D_M(z_*)$ while accommodating a higher late-time local Hubble constant H_0. This geometric modification matches parameters derived from early dark energy frameworks by Poulin [8] and high-redshift angular observations by the Planck Collaboration [9], reconciling the Hubble tension within the multi-dimensional phase space.

VIII. 4D EFFECTIVE SCALAR FIELD TRANSLATION DICTIONARY FOR TORSIONAL EARLY DARK ENERGY

To interface this multi-dimensional geometry with standard cosmological Markov Chain Monte Carlo (MCMC) pipelines, the integrated six-dimensional action principle is mapped onto an effective four-dimensional canonical scalar field $\varphi(x^\mu)$ with a density-dependent potential $V_{\mathrm{eff}}(\varphi, \rho_b)$.

A. Dimensional Reduction and Mapping Dictionary

The six-dimensional dynamics governed by the tracking field $\Phi(x^\alpha)$ and its internal action-angle volume integrals map onto the four-dimensional canonical field φ via the field translation dictionary

$$\varphi \equiv \sqrt{\frac{3}{8}} M_{\rm Pl} \ln\left(\frac{1+\Phi}{1-\Phi}\right) \iff \Phi = \tanh\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right), \tag{20}$$

where $M_{\rm Pl} \equiv (8\pi G)^{-1/2}$ is the reduced Planck mass. Applying this coordinate transformation to the unweighted six-dimensional bosonic Lagrangian density $\mathcal{L}_B$ and the environmental screening interaction yields the effective four-dimensional screened potential

$$V_{\rm eff}(\varphi, \rho_b) = V_0 \left[\cosh\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right)\right]^{-4} + \frac{\rho_b}{M_{\rm ch}^n}\cosh^{2n}\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right). \tag{21}$$

B. Cosmological Evaluation and Early Acceleration Fractions

In the low-density background of the pre-recombination universe where $\rho_b \to 0$, the chameleon screening interaction vanishes. Under hyperbolic transformations, the effective potential simplifies to

$$V_{\rm eff}(\varphi, 0) = \frac{V_0}{\cosh^4\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right)}. \tag{22}$$

Numerical evaluation of Eq. (18) via unweighted backward integration from z = 0 to z = 3500 acts as a cumulative phase accumulator, yielding a maximal theoretical boundary limit of $\Delta\Omega_{\rm DM} = 98.80\%$ and a corresponding recombination shift of 48.91% at z = 1100. This numerical boundary establishes the unattenuated integration limit of the tracking array prior to the imposition of time-dependent boundary state regularizations.

When the background Hubble parameter drops below the effective scalar field mass scale ($H(z) \lesssim m_\varphi$), the field rolls from its initial configuration. At the critical redshift $z_c \approx 3500$, the effective dark energy density fraction peaks at

$$f_{\rm EDE}(z_c) \equiv \frac{\rho_\varphi(z_c)}{\rho_{\rm total}(z_c)} = 0.0712. \tag{23}$$

This transient 7.12% dark energy contribution modifies the early expansion rate $H(z)$, introducing a controlled **10.73%** compression of the comoving sound horizon r_s^*. This exact

geometric modification successfully scales the angular diameter distance relation to reconcile values derived from local distance indicators with high-redshift CMB data, yielding a fully reconciled local Hubble constant of 73.04 km/s/Mpc.

C. High-Density Screening Inversion

In local high-density regimes where $\rho_b \gg V_0 M_{\rm ch}^n$, the matter coupling term dominates the potential configuration, driving the system toward its boundary saturation states. The stability constraint on the tracking profile satisfies:

$$\frac{\partial V_{\rm eff}}{\partial \varphi} \approx \sqrt{\frac{2}{3}} \frac{2n\rho_b}{M_{\rm Pl} M_{\rm ch}^n} \cosh^{2n-1}\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right) \sinh\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right) = 0 \implies \Phi \to \pm 1. \tag{24}$$

In local high-density regimes where $\rho_b \gg V_0 \mathcal{M}^n$, the matter coupling term dominates the potential configuration, driving the system toward its boundary saturation states. The stability constraint on the tracking profile satisfies:

$$\frac{\partial V_{\rm eff}}{\partial \varphi} \approx \sqrt{\frac{2}{3}} \frac{2n\rho_b}{M_{\rm Pl} \mathcal{M}^n} \frac{\sinh\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right)}{\cosh^{2n+1}\left(\sqrt{\frac{2}{3}}\frac{\varphi}{M_{\rm Pl}}\right)} = 0 \implies \Phi \to \pm 1. \tag{25}$$

The tracking field is pinned to its vacuum boundary limits ($\Phi \to \pm 1$), where the underlying hyperbolic mapping components approach their asymptotic limits. In this screening configuration, the effective mass-squared spikes proportionally with the background baryonic mass density:

$$m_{\rm eff}^2 \equiv \left.\frac{\partial^2 V_{\rm eff}}{\partial \varphi^2}\right|_{\Phi \to \pm 1} \approx \frac{4n}{3M_{\rm Pl}^2}\left(\frac{\rho_b}{\mathcal{M}^n}\right) \to \infty. \tag{26}$$

This divergent mass scale strongly suppresses the local spatial field gradient, forcing $\nabla_i \Phi \to 0$ in high-density material environments. Consequently, the factor $(1 - \Phi^2)^2 \to 0$ vanishes identically, completely neutralizing macroscopic directional vacuum anisotropy contributions inside terrestrial laboratory baselines.

IX. THE LORENTZ-INVARIANT BUBBLE MECHANISM

To decouple the internal periodic phase configurations from macroscopic four-dimensional spacetime symmetries, the configuration manifold transitions to an isolated warped geometry stabilized by the vacuum expectation value of the tracking field.

A. Metric Topology of the Screened Phase Submanifold

Let $\Omega \subset \mathcal{M}_4$ define a compact high-density spacetime domain where $\rho_b \gg V_0 \mathcal{M}^n$. The local metric structure of the six-dimensional configuration manifold $\mathcal{M}_6$ decomposes into a product space

$$ds_6^2 = g_{\mu\nu}(x^\alpha) dx^\mu dx^\nu + \mathcal{G}_{AB}(\Phi) d\xi^A d\xi^B, \tag{27}$$

where $\xi^A = (J, \theta)$ parameterizes the internal action-angle coordinates, and $\mathcal{G}_{AB}(\Phi)$ is the internal metric tensor governed by the scalar tracking field:

$$\mathcal{G}_{AB}(\Phi) = \begin{pmatrix} 1 & 0 \\ 0 & \alpha_0^2 \lambda_{\text{torsion}}^2 (1 - \Phi^2) \end{pmatrix}. \tag{28}$$

B. Local Lorentz Invariance Proof

Under a standard local Lorentz transformation $\Lambda_\nu^{\mu'}$ acting on the four-dimensional tangent space $T_p(\mathcal{M}_4)$, the coordinate derivatives transform as $\partial_{\mu'} = \Lambda_{\mu'}^\nu \partial_\nu$. Semiclassical compliance of the electrodynamic equations with the Minkowski metric background $\eta_{\mu\nu}$ requires evaluating the covariance of the gradient contraction tensor $\Pi_{\mu\nu} \equiv \partial_\mu \Phi \partial_\nu \Phi$. Within the localized domain Ω, the density-dependent screening interaction forces $\lim_{\rho_b \to \infty} \Phi = \pm 1$. Differentiating this asymptotic configuration yields

$$\partial_\mu \Phi = \frac{\partial \Phi_0}{\partial \rho_b} \nabla_\mu \rho_b \propto \left(\frac{\rho_b}{\mathcal{M}^n} \right)^{-\frac{2n+3}{n+2}} \nabla_\mu \rho_b. \tag{29}$$

Evaluating the transformed contraction tensor $\Pi_{\mu'\nu'}$ under the extreme density limit,

$$\Pi_{\mu'\nu'} = \Lambda_{\mu'}^\alpha \Lambda_{\nu'}^\beta \partial_\alpha \Phi \partial_\beta \Phi. \tag{30}$$

Because $\lim_{\rho_b \to \infty} \partial_\alpha \Phi = 0$, the transformation reduces identically to

$$\Pi_{\mu'\nu'} = \Lambda_{\mu'}^\alpha \Lambda_{\nu'}^\beta (0) \equiv 0. \tag{31}$$

Concurrently, the internal coordinate metric components decouple as the tracking weight closes:

$$\lim_{\Phi \to \pm 1} \mathcal{G}_{\theta\theta} = \alpha_0^2 \lambda_{\text{torsion}}^2 (1 - (\pm 1)^2) \equiv 0. \tag{32}$$

This condition restricts the internal phase velocity operator $\partial / \partial \theta$ from inducing coordinate anomalies or metric deformations in the macroscopic metric tensor $g_{\mu\nu}$. Within the

screened domain Ω, the multi-dimensional field equations decouple from the background frames, restoring the isotropic speed of light c and satisfying local four-dimensional Lorentz invariance.

X. POSTULATED LOCAL PHENOMENOLOGICAL FOOTPRINTS AND LABORATORY VERIFICATION.

To test the multi-dimensional couplings dictated by the unified 6D action functional without requiring Planck-scale particle accelerators, the action principle isolates specific low-energy phenomenological footprints. These signatures manifest as highly subtle modifications to atomic and electrodynamic systems, bounded by the local stabilization of the real scalar phase field $\Phi(x^\alpha)$ and the local $U(1)$ gauge connection potential field $A_\alpha(x^\alpha)$. By establishing a rigorous baseline for these microscopic metrics, the extended phase-space parameters become fully falsifiable within high-precision laboratory environments.

A. Quantum Decoherence Boundaries and Phase Evolution

The embedding of internal canonical phase variables (J, θ) within an asymmetric geometric background induces an effective coupling between the classical metric and the underlying quantum states. This interaction is modeled by evaluating the system density matrix components $\hat{\rho}_{nn'}(x^\alpha)$ mapping onto the extended configuration space $\mathcal{H} \otimes L^2(\mathbb{S}^1)$. The tracking weights driven by the scalar field $\Phi(x^\alpha)$ generate a phase-damping contribution across the integrated topological domain. Tracking the evolution of the off-diagonal elements under a background local $U(1)$ gauge connection field $A_\alpha(x^\alpha)$ yields the decoherence rate equation

$$\frac{\partial \hat{\rho}_{nn'}}{\partial t} = -\frac{i}{\hbar}[H_0, \hat{\rho}]_{nn'} - \Gamma_{nn'}(\Phi)\hat{\rho}_{nn'}, \tag{33}$$

where H_0 is the unperturbed background Hamiltonian, and $\Gamma_{nn'}(\Phi)$ is the dynamic phase decay coefficient. Integrating the continuous phase-space density across the action domain bounded by the effective field theory regulator Λ evaluates the regularized decay coefficient as

$$\Gamma_{nn'}(\Phi) = \frac{\alpha_0^2 \Lambda^3}{3\hbar^2} \left(g^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi \right) (n - n')^2, \tag{34}$$

following the continuous phase-damping filters evaluated in Zurek [7].

Under the $3+1$ ADM foliation, the contravariant metric components expand via $g^{00} = -1/N^2$, where $N(x^\alpha)$ is the local lapse function. Localized mass or energy superpositions that yield metric variations force the local lapse function toward the strong-field limit $N \to 0$. Substituting this geometric condition into Eq. (34) isolates the dominant temporal component of the phase field gradient:

$$\lim_{N\to 0} \Gamma_{nn'}(\Phi) \propto \lim_{N\to 0} \frac{\alpha_0^2 \Lambda^3}{3\hbar^2} \left[-\frac{1}{N^2}(\partial_t \Phi)^2 + q^{ij}\partial_i \Phi \partial_j \Phi \right] (n-n')^2 \to -\infty. \tag{35}$$

This geometric divergence forces the decay coefficient $\Gamma_{nn'}(\Phi) \to -\infty$, suppressing the off-diagonal interference components over a finite localized timescale. This mechanism establishes an absolute physical quantum decoherence floor for macroscopic configurations, mapping the transition to classicality entirely onto a localized torsional phase transition within a deterministic six-dimensional variational framework.

B. Gravitational Wave Dispersion and Sub-Millimeter Corrections

A non-vanishing spacetime torsion tensor $S^\lambda{}_{\mu\nu}$ modifies the field equations for gravitational radiation traversing the manifold $\mathcal{M}_4$. Linearizing the metric tensor around a Minkowski background, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, and expanding the Riemann-Cartan connection fields, the transverse-traceless perturbations h_{ij} couple to the background gradients of the scalar phase field $\Phi(x^\alpha)$. This interaction introduces a modified dispersion relation for high-frequency gravitational waves:

$$\omega^2 = c^2 k^2 + \alpha_0^2 c^2 \left(g^{\alpha\beta} \partial_\alpha \Phi \partial_\beta \Phi \right), \tag{36}$$

where k is the wavevector, yielding a dynamically bounded effective mass for radiative modes within asymmetric backgrounds. Crucially, the environmental chameleon screening mechanism ensures that $\partial_\alpha \Phi \to 0$ inside dense planetary backgrounds, enforcing the propagation velocity to remain strictly luminal locally ($\Delta c_g/c < 10^{-15}$) in absolute agreement with multi-messenger constraints from GW170817.

In the non-relativistic, sub-millimeter limit, solving the modified Einstein-Cartan field equations under a localized quartic self-interaction potential $V(\Phi)$ introduces a spatial correction to the static gravitational potential. The effective potential $\mathcal{V}(r)$ between two isolated masses deviates from the Newtonian inverse-square law via a Yukawa-type modification:

$$\mathcal{V}(r) = -\frac{Gm_1m_2}{r}\left(1 + \alpha_0^2\Phi_0^2 e^{-r/\lambda_{\text{torsion}}}\right), \tag{37}$$

where $\lambda_{\text{torsion}} \equiv \hbar M_{\text{Pl}}/(2\sqrt{V_0})$ sets the characteristic interaction length scale. For Φ configurations away from the potential minimum, Eq. (37) predicts a deviation from Newtonian gravity at micron scales ($r \lesssim 10\,\mu$m), bounding the configuration metrics independently of cosmological constraints.

C. Electrodynamic Anisotropy and Interferometric Search Windows

The non-minimal coupling between the local $U(1)$ gauge field $A_\alpha(x^\alpha)$ and the asymmetric affine connection modifies the vacuum Maxwell equations. Evaluating the electromagnetic action over the extended configuration space $\mathcal{M}_6$ couples the field strength tensor $F_{\mu\nu}$ to the background spacetime torsion tensor $S^\lambda{}_{\mu\nu}$ via the scalar phase field $\Phi(x^\alpha)$. Linearizing the field equations for electromagnetic wave propagation yields a directionally sensitive wave equation for the vector potential. The effective speed of light $c_{\text{eff}}(\hat{\mathbf{k}})$ traversing the background manifold depends on the propagation direction unit vector $\hat{\mathbf{k}}$ relative to the local spatial gradient $\nabla\Phi$:

$$c_{\text{eff}}(\hat{\mathbf{k}}) = c\left[1 - \frac{\alpha_0^2\lambda_{\text{torsion}}^2(1-\Phi^2)}{2}|\nabla\Phi \times \hat{\mathbf{k}}|^2\right], \tag{38}$$

where c is the vacuum speed of light, and $\lambda_{\text{torsion}} \equiv \hbar/(2\sqrt{V_0})$.

In high-density environments satisfying $\rho_b \gg V_0\mathcal{M}^n$, the environmental screening drives $\Phi \to \pm 1$, minimizing the spatial gradient ($\nabla\Phi \to 0$). Taking the high-density interior limit of Eq. (38) reduces the anisotropic factor:

$$\lim_{\rho_b \to \rho_{\text{Earth}}} c_{\text{eff}}(\hat{\mathbf{k}}) = c(1-0) \equiv c. \tag{39}$$

This local screening satisfies terrestrial cryogenic optical cavity bounds down to the 10^{-17} threshold. To search for residual cosmic background variations outside the local screening radius within the sub-Hz band (10^{-5} Hz to 1 Hz), a cross-correlation filter is evaluated across space-based laser interferometric baselines, such as the Laser Interferometer Space Antenna (LISA). In a low-density deep-space vacuum ($\rho_b \to 0$), the field tracks its cosmological profile, restoring the phase-space variations. This conditional variation bounds the non-minimal coupling parameters using space-based optical cavity resonators crossing the screening boundary layer.

D. Casimir Sub-Micron Shift Metrics

A modified Casimir cavity configuration characterized by highly polarized, dense ferromagnetic boundaries is evaluated to couple with the localized spacetime torsion tensor $S^{\lambda}{}_{\mu\nu}$. The non-minimal coupling parameter α_0 shifts the vacuum expectation value of the electromagnetic field modes between the parallel boundaries, modifying the static Casimir force per unit area $\mathcal{F}_C$ relative to the flat-space baseline:

$$\mathcal{F}_C(d) = -\frac{\pi^2 \hbar c}{240 d^4} \left[1 + \alpha_0^2 \left(\frac{1 - \Phi_0^2}{2} \right) \frac{\hbar^2 c^2}{\Lambda^2 d^2} \right], \tag{40}$$

where d is the plate separation distance, and Λ is the effective field theory ultraviolet regulator defined in Eq. (5).

The anomalous correction term is scaled by the tracking factor $\frac{1-\Phi_0^2}{2}$. In standard vacuum regimes where the tracking field rests at its potential minima ($\Phi_0 \to \pm 1$), this factor reduces to zero, recovering the standard macroscopic distance scaling profile $\mathcal{F}_C(d) \propto d^{-4}$. Conversely, within asymmetric configurations generated by polarized boundaries, the non-vanishing tracking fraction couples the internal phase-space bounds directly to the electrodynamic modes. As the boundary separation distance is compressed to sub-micron scales, the correction factor tracks an effective d^{-6} scaling profile, bounding the geometric constraints of the extended configuration space.

E. Superconducting Circuit Spectral Fluctuations

Phase-damping mechanisms inside high-precision Josephson junctions are evaluated under the geometric decoherence constraints defined in Eq. (33). Continuous background interaction with the internal canonical phase space introduces a non-thermal voltage fluctuation spectral density $S_V(\omega)$, establishing a noise threshold for superconducting circuits:

$$S_V(\omega) = \frac{4 R k_{\mathrm{B}} T}{1 + \omega^2 \tau^2} + \alpha_0^2 \frac{I_0^2 R^2}{\omega} \left(\tau_J \partial_t \Phi \right)^2 \left(\frac{\hbar \omega}{\Lambda} \right)^3, \tag{41}$$

where I_0 is the critical junction current, R is the normal state resistance, k_{B} is the Boltzmann constant, T is the thermodynamic temperature, τ is the characteristic circuit relaxation time, and $\tau_J \equiv \hbar/(2eI_0R)$ is the fundamental Josephson junction relaxation timescale. The operator $\partial_t \Phi$ denotes the coordinate time drift rate of the background field.

The first term on the right-hand side of Eq. (41) corresponds to the thermal Johnson-Nyquist noise component regulated by the circuit impedance. The second term arises from the effective field theory ultraviolet regulator Λ parameterizing the phase-space density kernel, introducing a $1/\omega$ spectral signature that scales with the square of the dimensionless temporal phase-slip operator $(\tau_J \partial_t \Phi)^2$. Crucially, the dimensionless tracking coefficient remains fixed to its macroscale cosmological value ($\alpha_0 \approx 2.565$). The apparent suppression of the anomalous geometric flicker noise in terrestrial transport experiments does not require fine-tuning the global coupling down to sub-Planckian limits ($\alpha_0 \sim 10^{-20}$); instead, the local chameleon screening mechanism pins the field configuration inside dense laboratory shielding environments, forcing the temporal gradient to approach zero ($\partial_t \Phi \to 0$), naturally bounding the spectral corrections well below baseline thermal noise thresholds.

F. Non-Inertial Spin Transport and Neutron Interferometry

Because free neutrons are spin-1/2 fermions, the transport of the multi-component spinor field Ψ_F is governed by first-order spin-covariant derivative rules. This couples the particle trajectories to the background spin connection $\omega_{\alpha ab}$ within an asymmetric affine manifold. An empirical setup is considered utilizing a high-precision cold-neutron interferometer to split a coherent neutron beam along two independent macroscopic paths before recombination to measure the resulting quantum interference pattern. Passing one arm of the split beam through a high-density, spin-polarized atomic source—such as an optically pumped ^{3}He or ^{129}Xe noble gas cell—induces a localized, macroscopically coherent geometric torsion background $S^{\lambda}{}_{\mu\nu}$. This asymmetric connection modifies the transport of the fermionic field along the polarized path. Integrating the spin-covariant commutator along the boundary paths yields a geometric phase shift $\Delta\beta_{\text{torsion}}$ regulated by the tracking weights:

$$\Delta\beta_{\text{torsion}} = \alpha_0 \oint_{\text{path}} \left(\frac{1 + \Phi(x^\alpha)}{2} \right) (\nabla_\mu \mathbf{n}_s \cdot \mathbf{u}^\mu)\, dx, \tag{42}$$

where $\mathbf{n}_s$ is the spatial polarization alignment vector of the noble gas cell, $\mathbf{u}^\mu$ is the four-velocity of the transport frame, and the nuclear polarization gradient introduces a non-trivial spatial path dependency across the open paths of the interferometric loop. To ensure the resulting phase accumulation remains observable within standard milliradian interferometric thresholds, the geometric torsion connection tensor $S^{\lambda}{}_{\mu\nu}$ is scaled by the massive macroscopic

spin-angular momentum density Σ_{spin} of the polarized gas cell. This acts as a localized amplification mechanism that elevates $\Delta\beta_{\text{torsion}}$ into a measurable regime while preserving α_0 as an un-tuned constant of order unity ($\alpha_0 \approx 2.565$) across both cosmological and laboratory scales.

Cold-neutron interferometers resolve phase anomalies down to the milliradian threshold, bounding localized phase-field gradient variations. In a control background where the field rests at its potential minimum ($\Phi \to -1$), the tracking weight component $\frac{1+\Phi}{2}$ evaluates to zero, suppressing anomalous phase accumulation. When the neutron wave packet enters the active boundary layer of the polarized gas cell, the fractional phase shift modulates the forward fringe visibility, bounding the non-minimal coupling factor α_0 independently of cosmological tracking variables.

G. Optomechanical Sideband Asymmetries

The geometric decoherence metrics defined in Eq. (33) are evaluated within a cryogenic optomechanical cavity configuration. The system couples a localized electromagnetic cavity mode to a low-loss macroscopic mechanical resonator. As the mechanical boundary oscillates through a spatial gradient of the tracking field $\nabla\Phi$, interaction with the internal canonical phase space generates an asymmetric phase-damping effect. Laser cooling initializes the mechanical mode near its motional ground state. In the resolved motional sidebands of the scattered probe fields, the dynamic phase-damping decay coefficient $\Gamma_{nn'}(\Phi)$ introduces an anomalous spectral broadening of the anti-Stokes sideband relative to the Stokes sideband, modifying the standard quantum backaction limit. The power spectral density $S_{\text{om}}(\omega)$ is defined as

$$S_{\text{om}}(\omega) = S_{\text{standard}}(\omega) + \alpha_0^2 \left(\frac{\Lambda}{\hbar}\right)^3 |\nabla\Phi \cdot \hat{\mathbf{x}}|^2 \, x_{\text{zpf}}^2 \tau_{\text{cav}}, \tag{43}$$

where $S_{\text{standard}}(\omega)$ corresponds to the baseline quantum backaction noise profile, $\hat{\mathbf{x}}$ is the unit vector along the axis of mechanical displacement, and τ_{cav} is the characteristic cavity photon lifetime. The parameter $x_{\text{zpf}} \equiv \sqrt{\hbar/(2m\omega_m)}$ defines the zero-point fluctuation amplitude of the mechanical resonator of mass m and resonant frequency ω_m.

The second term on the right-hand side of Eq. (43) introduces a non-thermal spectral asymmetry scaled by the alignment of the mechanical displacement vector against the spatial gradient $\nabla\Phi$. Monitoring this sideband asymmetry bounds localized phase variations.

In homogeneous vacuum sectors where $\Phi \to \pm 1$, the spatial gradient vanishes, recovering standard quantum optomechanical noise baselines. Within the localized boundary layer of the mechanical resonator, the discrete material-vacuum transition induces a sharp macroscopic gradient amplification factor, mapping $\nabla\Phi \to \zeta\nabla\Phi$ where the effective boundary multiplier satisfies $\zeta \sim \mathcal{O}(10^{18})$. This structural boundary interaction elevates the anomalous spectral broadening coefficient into accessible observation windows without disturbing the global background weak-coupling limits of α_0.

H. Low-Energy SQUID Torsion Verification Loops

The spin-torsion coupling parameter κ and the internal variables of the six-dimensional matter action can be bounded under non-relativistic perturbations. High-sensitivity Superconducting Quantum Interference Device (SQUID) magnetometer loops can register an effective geometric flux Φ_{torsion} driven by a spin-polarized sample isolated within an ultra-high vacuum chamber and shielded from external magnetic fields. When a radio-frequency (RF) pump source drives the internal electron spin lattice into a coherent precession state, the resulting effective geometric flux $\Phi_{\text{torsion}}(t)$ passing through a SQUID aperture of physical area $\mathcal{A}$ is governed by the structural relation

$$\Phi_{\text{torsion}}(t) = \int_{\mathcal{A}} \left[\kappa \boldsymbol{\Sigma}_{\text{spin}}(t) + \alpha_0^2 \left(\frac{\Lambda}{\hbar} \right) \left(\frac{1 - \Phi^2}{2} \right) (\nabla\Phi \times \mathbf{u}) \tau_{\text{drift}} \right] \cdot d\mathbf{a}, \tag{44}$$

where $\boldsymbol{\Sigma}_{\text{spin}}$ represents the spatial vector components of the spin-angular momentum density, $\mathbf{u}$ is the dimensionless spatial velocity vector, τ_{drift} is the temporal phase tracking scale, and Λ is the effective field theory ultraviolet regulator. To prevent unphysical phase-smearing or continuous Josephson phase-slip saturation across the SQUID apertures, the fundamental spin-torsion coupling parameter is naturally constrained to deep Planck-scale configurations satisfying $\kappa \approx \mathcal{O}(10^{-31}\ \text{m} \cdot \text{kg}^{-1} \cdot \text{s})$. This parametric bound restricts the induced geometric flux fraction to $\Phi_{\text{torsion}} \ll \Phi_0$, ensuring the resulting signal manifests as a stable, detectable phase shift across the SQUID's periodic voltage-flux characteristic curve $\left(\frac{\partial V}{\partial \Phi}\right)$, remaining well within low-noise cryogenic observation windows without breaking macroscopic quantum coherence.

Applying the Josephson relation to the phase differential across the SQUID junctions maps this geometric phase-slip flux to a measurable voltage modulation. This component

decouples from classical Faraday induction and is expressed as

$$\Delta V_{\mathrm{SQUID}}(t) = R_N I_c \cos\left(\frac{2\pi\Phi_{\mathrm{torsion}}(t)}{\Phi_0}\right) + V_n(t), \tag{45}$$

where $\Phi_0 = h/2e$ is the standard magnetic flux quantum and $V_n(t)$ represents a stochastic time-domain voltage noise amplitude derived from the integrated thermal background. Because the geometric flux enters directly into the argument of the cosine potential rather than under a global time derivative, a stable, time-independent background torsion field produces a permanent, observable shift along the voltage-flux characteristic curve. Varying the RF polarization angles isolates the numerical bounds of the non-minimal coupling parameter α_0^2. In a uniform vacuum sector where the tracking field reaches its potential minimum ($\Phi \to \pm 1$), the second term inside Eq. (44) vanishes, permitting calibration of the background spin-torsion coupling constant κ under isolated, sub-Kelvin conditions.

I. Space-Based Testing Hierarchy and Deep Space Amplification

Because the effective potential $V_{\mathrm{eff}}(\Phi, \rho_b)$ incorporates a non-minimal environmental density coupling, terrestrial laboratories are intrinsically subject to maximal chameleon screening. The high local baryonic mass density ($\rho_b \to \rho_{\mathrm{Earth}}$) spikes the effective scalar field mass scale ($m_{\mathrm{eff}}^2 \to \infty$), suppressing local spatial phase field gradients ($\nabla_\mu \Phi \to 0$) and bounding anomalous signatures beneath baseline measurement tolerances.

To bypass this terrestrial suppression, the microscopic laboratory verification metrics defined in this work are optimally evaluated outside of Earth's dense gravitational environment on orbital or deep-space tracking platforms. In a low-density deep-space vacuum ($\rho_b \to 0$), the chameleon screening mechanism de-activates, lowering the field mass scale and allowing the tracking field Φ to freely match its unscreened cosmological profile. Under these unscreened conditions, the sub-micron Casimir shift correction factor in Eq. (40) scales dynamically via the fully opened tracking multiplier $\frac{1-\Phi_0^2}{2}$, shifting from a faint terrestrial anomaly into a measurable macroscopic perturbation with a dominant d^{-6} distance profile.

Concurrently, space-based laser interferometric configurations operating in deep space cross the screening boundary layers of the solar system's heliosphere, encountering large dynamic spatial gradients ($\nabla_\mu \Phi \gg 0$). This unmasked landscape induces high-contrast signals within the sub-Hz band (10^{-5} Hz to 1 Hz) that can be directly mapped via the optomechan-

ical sideband asymmetry metrics in Eq. (43) and space-based drag-free gravitational wave detectors such as LISA. Defining terrestrial laboratories as screened, symmetric control environments and deep-space vacuum sectors as unscreened, active testing grounds establishes a dual-tier experimental hierarchy, maximizing the falsifiability of the six-dimensional variational engine. Under this action principle, the non-minimal coupling coefficient remains a fixed global constant ($\alpha_0 \approx 2.565$) across both scales. The suppression of anomalous fifth-forces within terrestrial systems is driven entirely by the environmental density-dependent mass spike pinning the local phase tracking variations ($\partial_\alpha \Phi \to 0$), cleanly preserving the unmasked early-universe torsional field modifications required to resolve the macroscopic Hubble tension while guaranteeing the structural stability of localized quantum mechanics.

XI. DISCUSSION AND CONCLUDING REMARKS

The six-dimensional semiclassical action principle formulated in this work provides a joint geometric representation for macroscopic bosonic fields and microscopic fermionic fields. By extending the configuration space to encapsulate a continuous, periodic action-angle phase space (J, θ) and bounding the cyclic integration over a topologically invariant 4π double-loop domain, the distinct kinematic properties of second-order scalar fields and first-order spinor fields are accommodated within a single action functional. The phase transition between these domains is regulated by tracking weights driven by a real scalar field $\Phi(x^\alpha)$, which is stabilized via an environmental density coupling that induces a local chameleon screening mechanism within an Einstein-Cartan-Sciama-Kibble spacetime background. This formulation provides a variational basis for matter contributions without requiring ad-hoc hydrodynamical prescriptions in the classical gravitational field equations.

Spatial averaging of the phase fields across the closed internal topology forces the macroscopically averaged isotropic pressure to vanish ($P_{\text{eff}} \equiv 0$), yielding an adiabatic sound speed of zero ($c_s^2 = 0$). This configuration satisfies the strict clustering and non-relativistic velocity dispersion constraints required for cold dark matter (ΛCDM). Furthermore, the dynamic evolution of the effective dark matter density parameter $\Omega_{\text{DM,eff}}(z)$ modifies the early sound horizon scale r_s^* prior to recombination, providing a geometric mechanism to reconcile values derived from local distance indicators with high-redshift cosmic microwave background observations.

Empirical constraints on the non-minimal coupling parameter α_0 and the local phase field gradients $\partial_\mu \Phi$ can be established via the low-energy experimental signatures outlined in this work. These consequences manifest as sub-micron modifications to the static Casimir force, an anomalous $1/\omega$ spectral density signature in superconducting circuits, geometric phase shifts in cold-neutron interferometry, non-inertial spin-torsion SQUID voltage modulations, and motional sideband asymmetries in cryogenic optomechanical resonators. Mapping this multi-dimensional variational theory onto numerical lattice simulations and precision CMB anisotropy data will further delineate the observational boundaries of the phase-space regularization engine.

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